



Stochastic Modeling of Glaciar Melting via Monte Carlo Simulations.

Course: CSE400 – Fundamentals of Probability in Computing
Domain: Climate

GROUP-12

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Mathematical Problem Formula

- Deterministic to stochastic: Used Probability density function(PDFs) to model climate variability.
- Equation:
$$M = \int [P(t) \cdot S(T_a) - E(T_a, \alpha_i, n)] dt.$$
- Defining glacial mass balance as a “Transfer Function”.
- Inputs are random variable and the output is a joint PDF.

Air Temperature(T_a)

- PDF Model: Gaussian Distribution $\mathcal{N}(0, 0.424^2)$
- Mathematical Support: Defining the interval where the PDF is non-zero.
- Chebyshev proof: Using $\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$ to define the 3σ “credible bound” of $\pm 1.27^\circ C$.

Precipitation (P) – Skewed Distribution Theory

- PDF Model: Gamma Distribution $f(x; k, \theta)$ to ensure non-negativity ($P \geq 0$)
- Markov inequality application: $\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$
- Physical Constraint: Defining the range from 0 to positive infinity to prevent "negative rain".



Ice Albedo (α_i)– Truncated Support

- PDF Model: Truncated Normal Distribution restricted to $[0.2, 0.5]$.
- Analytical Logic: Standard Gaussian tails are physically impossible for reflectivity ratios (must be $0 < \alpha < 1$).
- Parameterization: Normalizing the PDF constant to ensure the total area equals 1 over the restricted range.

Snow Threshold - The Transition Curve

- Formula: $P(\text{snow}) = 1 - \Phi\left(\frac{T - \mu}{\sigma}\right)$
- Terms: $\mu = 1.5^{\circ}\text{C}$ (*ThresholdCenter*), $\sigma = 0.5^{\circ}\text{C}$ (*UncertaintyWidth*)
- Model Type: Cumulative Distribution Function (CDF) representing a probabilistic transition.
- Empirical Limits: *Mathematically focused on the transition zone of 0°C to 3°C .*

Cloudiness - Non-Linear Energy Mapping

- Representation: Logical sky fraction $n \in [0, 1]$
- Mapping Formula: $T_{cl} = 1.0 - 0.233n - 0.415n^2$
- Coefficients: Derived from radiation studies showing non-linear sun-blocking effects
- Jensen Inequality: Proves that the average of the function is not equal to the function of the average.
- Support: Logical Limit of $[0, 1]$

The Solver- Surface Energy Balance (SEB)

- Master Equation: $Q_{melt} = S_{in}(1 - \alpha)T_{cl} + L_{in} + Q_h + Q_l$
- Three-Stage Physical Melt Chain:
- Stage 1: Melt Energy Equation.
- Stage 2: Melt Depth Conversion (Latent Heat of Fusion).
- Stage 3: Seasonal Melt Magnitude aggregation.
- Method: Numerical Integration on paper to resolve non-linear interactions.

Theoretical Stability- Law of Large Numbers

- Theorem: Weak Law of Large Numbers (WLLN).
- Core Principle: Sample mean converges to true mean as $n \rightarrow \infty$
- Stability Math: Variance of estimate reduces by $\frac{\sigma^2}{n}$
- Verification: Guarantees that the stochastic results settle into a stable, physical average.

Risk Quantification- Chebyshev's Inequality

- Theorem: $\mathbb{P}(|X - \mu| \geq \alpha) \leq \frac{\text{Var}(X)}{\alpha^2}$
- Utility: Provides "Safety Bounds" for the model without assuming a specific distribution shape.
- Resulting Output: Identification of the 95th percentile to quantify Extreme Melt Risk.
- Final Bound: Defines the rigorous upper and lower limits for high-confidence risk assessment.